



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
Higher 2

NAME

ANSWERS

INDEX NO

CG

BIOLOGY

9744/03

Paper 3 Long Structured and Free-Response Questions

13 Sep 2022

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and class in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

Indicate the question you have attempted at the top of page **18**.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	30
2	10
3	10
Section B	
4 or 5	25
Total	75

This document consists of **25** printed pages and **1** blank page.

Section A

Answer **all** questions in this section.

- 1 Estrogen receptors (ER) are expressed in mammary cells. Fig. 1.1 shows how an ER serves its functions at target cells:

- as an intracellular receptor activated by the steroid hormone estrogen (E), when E binds to complementary ligand binding site on ER,
- and as a transcriptional activator, after two ER bound with E dimerised and enters the nucleus.

Estrogens induce transcription of genes in their target cells. Such genes have estrogen response elements (ERE) as enhancers.

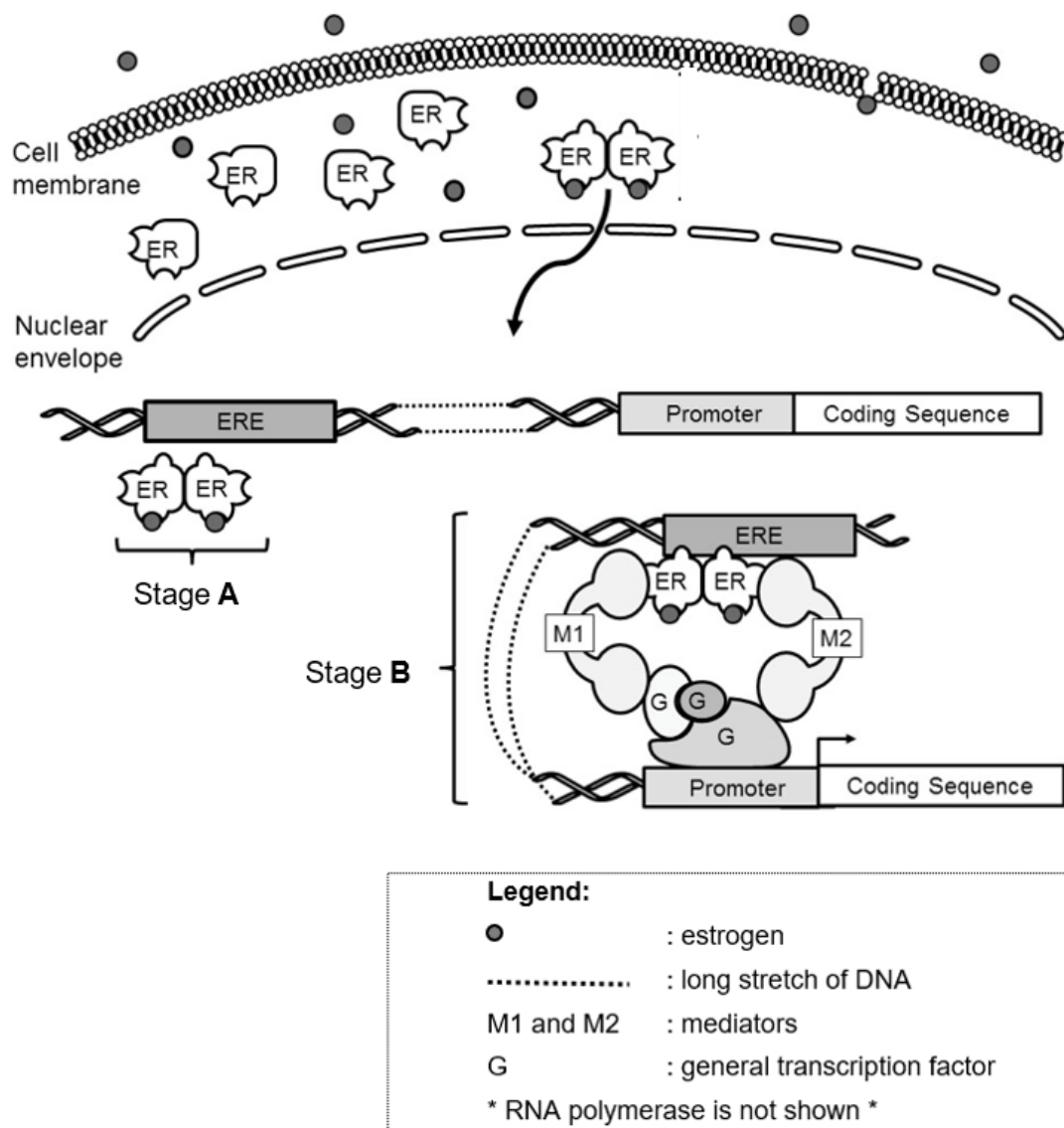


Fig. 1.1

(a) With reference to Fig. 1.1, describe stages **A** to **B**.

Stage A: 1. dimerized ER bound to E binds via complementary fit to ERE

acts as transcriptional activator;

Stage B: 2. E-ER dimer recruits DNA bending proteins resulting in DNA looping,

E-ER dimer brought closer to the promoter;

3. recruit mediators (M1 and M2)

and general transcription factors (G);

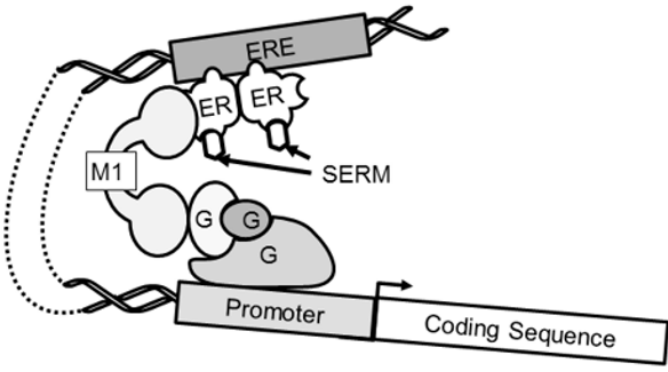
4. RNA polymerase binds (more stably) to promoter

forming transcription initiation complex leading to increased rate of transcription;

[4]

Estrogen receptors are over-expressed in around 70% of breast cancer cases. Research into the treatment of metastatic breast cancer focused on the use of drugs such as selective estrogen-receptor modulator (SERM) and estrogen receptor downregulator (ERDR) as shown in Table 1.1.

Table 1.1

drug	function of drug	presence of drug-receptor complex in the nucleus	effect of drug on the target gene
SERM ○		✓	
ERDR ★		✗	Basal rate of transcription

(b) With reference to Table 1.1,

(i) explain how SERMs can be used in the treatment of breast cancer.

1. SERMs are structurally similar to estrogen / E

(®: similar to ER / same shape as ER)

compete with E for binding with ER / binds to ER, blocking E from binding;

(®: binds to active site of ER)

2. SERMs recruit M1 but not M2 / does not recruit M2 / prevent complete recruitment / assembly of mediators / prevent complete DNA bending

formation of stable transcription initiation complex is reduced / prevented;

3. This reduces rate of oncogene / cancer critical gene expression OR

reduce rate of transcription of gene that result in excessive cell division / rate of cell division exceeding rate of cell death;

[3]

(ii) suggest why ERDRs might be more effective than SERMs.

1. ERDRs prevent movement of ERDR-ER complexes to the nucleus

/ ER degraded in the cytoplasm,

preventing increased oncogene expression / transcription;

2. SERMs still allows movement of ER into the nucleus / enables recruitment of M1 and general transcription factors /

transcription initiation complex to be formed for transcription / gene expression;

[2]

- (c) Cancer is described as a multi-step process. Explain how mutations may lead to uncontrolled proliferation in cells lining the milk ducts, leading to breast cancer.

1. loss of function mutation of tumour suppressor gene
/ mutations must affect both alleles of tumour suppressor gene,

non-functional tumour suppressor protein formed upon gene expression;

2. loss of arrest of cell division, loss of ability for DNA repair and loss of apoptosis;

3. gain of function mutation of proto-oncogene
/ only need one mutated allele;

hyperactive protein upon gene expression;

OR

3a. gain of function mutation in promoter of proto-oncogene,

excessive amount of proteins formed upon gene expression;

4. leads to excessive stimulation of the cell cycle / cell keeps dividing;

5. accumulation of other mutations in a single cell

resulting in uncontrolled proliferation of cells lining the milk ducts;

AVP e.g.:

6. activation of telomerase gene, enabling the maintenance / replacement of telomeres,

cells are able to overcome the Hayflick limit / continue dividing (despite presence of mutations);

7. activation of genes involved in angiogenesis,

blood vessels formed supply nutrients and oxygen to tumour cells (and remove waste);

8. Activation of genes involved in metastasis,

enabling cells to travel from primary tumour site to establish themselves in a different location in the body / migrate into neighbouring tissues;

OR

9. Increased expression of genes coding for extracellular proteases to degrade cytoskeleton,

promoting metastasis into nearby blood vessels / tissues;

[4]

Any 4 marking points

Fig. 1.2 shows that in breast cancer cells, there is an overexpression of human epidermal growth factor receptor 2 (HER2) proteins. HER2 functions as a receptor protein on the cell surface membrane to send a signal to the nucleus for the cell to undergo cell division.

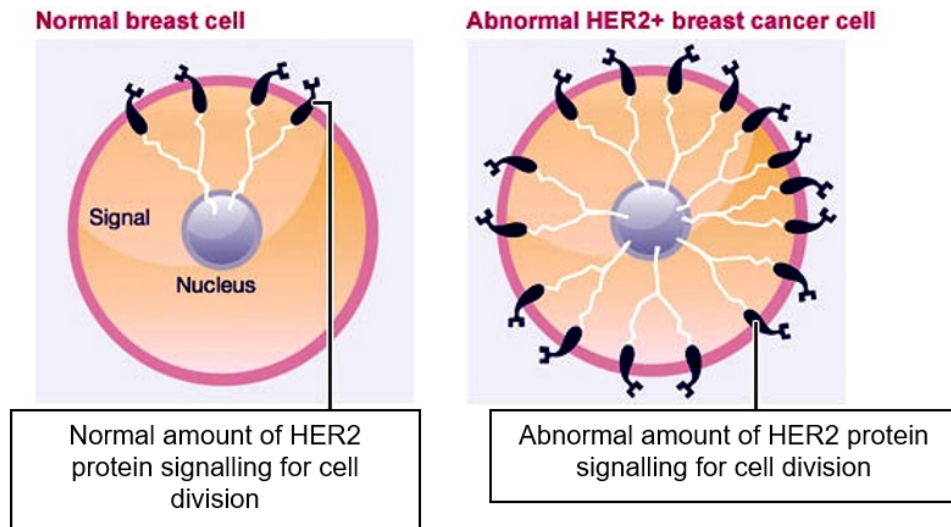


Fig. 1.2

Trastuzumab, an antibody specific to HER2, is an effective treatment for early-stage HER2+ breast cancer. However, advanced-stage HER2+ breast cancer patients tend to develop resistance to the treatment.

HER2+ breast cancer cells also overexpress CD47, a surface protein that helps cancer cells evade macrophage phagocytosis. Research has shown that blockage of CD47 using an antibody specific to CD47 (α -CD47 mAb) is also an effective treatment for HER2+ breast cancer.

An investigation was conducted to determine the effectiveness of treating HER2+ breast cancer cells with both trastuzumab and α -CD47 mAb.

Groups of genetically identical breast cancer cells were subjected to 3 different treatments:

1. incubation with α -CD47 mAb
2. incubation with trastuzumab
3. incubation with both trastuzumab and α -CD47 mAb

The investigation was done twice, with 3 replicates for each treatment.

The percentage of phagocytosis by macrophages was assessed and shown in Fig. 1.3.

Data are presented as mean $\pm$ S.D. (standard deviation).

Statistical differences between the two groups of cells are marked by asterisk ($P \leq 0.0001$).

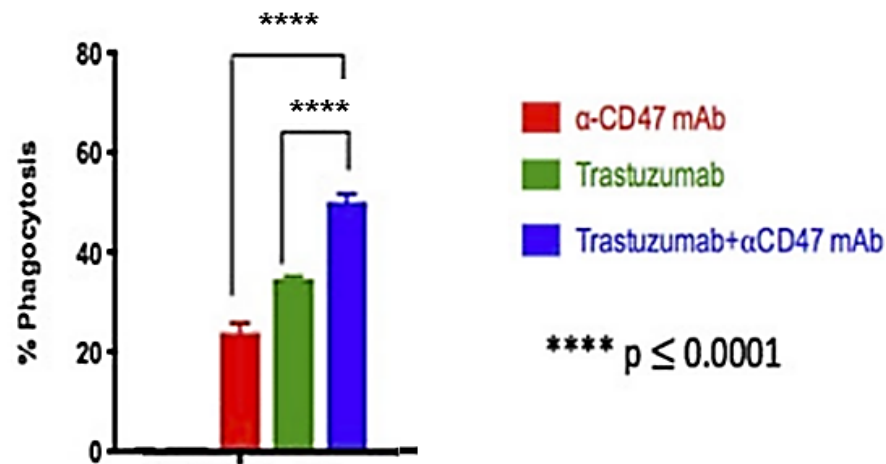


Fig. 1.3

- (d) Using Fig. 1.3 and the information provided, comment on the effectiveness of the combined treatment versus a treatment using either trastuzumab, or α-CD47 mAb alone in humans. Discuss on the confidence of the presented data.

1. Combined treatment with trastuzumab and α-CD47 mAb is more effective as shown by the higher % of phagocytosis

compared to trastuzumab / α-CD47 mAb alone;

2. [QV] e.g. % phagocytosis for trastuzumab and α-CD47 mAb treatment is higher at 50% (accept 48 - 52%) vs. 25% (accept 22 - 26%) for α-CD47 mAb alone;

OR

% phagocytosis for trastuzumab and α-CD47 mAb treatment is higher at 50% (accept 48 - 52%) vs. 37% (accept 35 - 38%) for trastuzumab alone;

There is confidence in the data due to:

3. High reproducibility / reliability from performing of a repeat experiment / 3 replicates done per experiment;

4. Ref. statistical test analyses revealed a very significant difference between treatment with α-CD47 mAb alone / with trastuzumab alone

vs combined treatment, hence results not due to chance alone;

5. Ref. relatively small standard deviation for all groups, suggesting consistency in results with respect to the mean;

There is lack of confidence in the efficacy of treatment due to:

6. Experiment performed on breast cancer cell line *in vitro*, need more clinical studies on humans;

7. Lack of a control group / untreated cells for better comparison and conclusions;

8. AVP;

[4]

Marking points 1 & 2 + AND 2 marking points from 3 to 8 for full credit

An early diagnosis of many types of cancer can result in successful treatment.

The BRCA2 protein is involved in suppressing the development of tumours.

Several different dominant alleles of the gene, *BRCA2*, code for faulty versions of the BRCA2 protein. The presence of any one of these faulty alleles leads to an increased chance of developing several types of cancer, including breast and lung cancers. Not everyone with one of these alleles develops cancer. This is because environmental factors, including lifestyle, are also involved.

Fig. 1.4 is a pedigree (family tree) showing the occurrence of cancers in four generations of a family. The presence of a faulty *BRCA2* allele was confirmed in person 15. The other individuals with cancer were not tested for the presence of the allele.

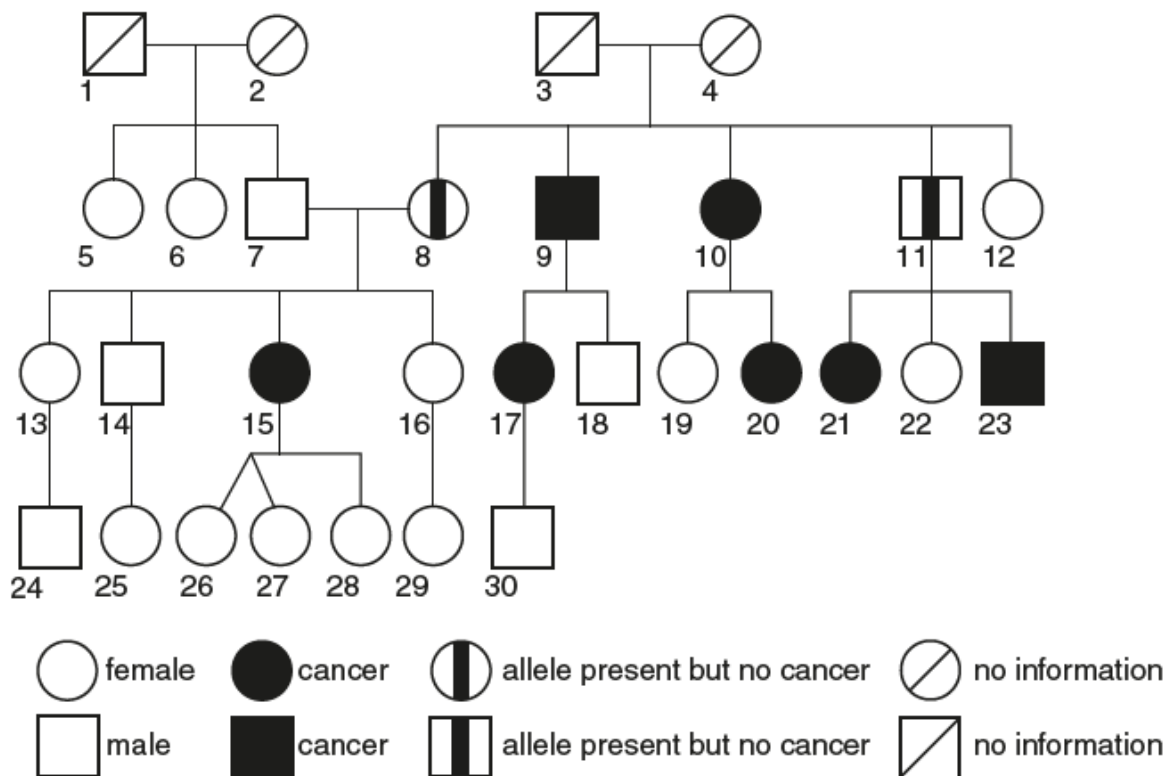


Fig. 1.4

- (e) Discuss the extent to which Fig. 1.4 provides evidence that a faulty *BRCA2* allele increases the risk of a person developing cancer.

Any 3 marking points below:

1. individual 8 or 11 has *BRCA2* allele, but does not have cancer;

2. no evidence / unknown, that individuals, apart from 15, with cancer have, *BRCA2* allele

OR

3. individuals with cancer, apart from 15, may have a different mutation;

4. no children of individual 15, are known to have the *BRCA2* allele / have cancer;

5. individuals in fourth generation / children of individual 15, may develop cancer later in life;

6. individual 15 has cancer and *BRCA2* allele;

7. (some) individuals with cancer in third generation had a parent with cancer but unknown to have *BRCA2* allele;

OR

8. some) individuals with cancer in third generation had a parent with *BRCA2* allele;

9. individual 3 or 4 may have had the *BRCA2* allele;

OR

10. any individual from 8 to 11 may have inherited the *BRCA2* allele, from individuals 3 or 4;

11. overall data is inconclusive;

[3]

- (f) Traditionally, stem cells from bone marrow have been used to treat patients with leukaemia, a cancer of the blood or bone marrow.

Recent studies have shown that stem cells taken from umbilical cord blood may be more effective in treating leukaemia than stem cells taken from bone marrow.

Table 1.2 shows the results from patients with leukaemia after treatment with stem cells from unrelated sources over a two-year period.

Table 1.2

source of stem cells	probability / %	
	leukaemia-free survival	death from immune rejection of transplant
cord blood	34	12
bone marrow	33	31

Using information from Table 1.2 and your knowledge on stem cells, suggest why stem cells from the cord blood may be more preferred in the treatment of leukaemia.

1. similar probability for survival

cord blood 34% vs bone marrow 33%;

2. lower probability of death from immune rejection

cord blood 12% vs bone marrow 31%;

3. idea of ease of obtaining e.g. easier to obtain cord blood from discarded umbilical cord / no need for invasive surgery

compared to invasive procedures to obtain bone marrow;

OR

4. idea of storage e.g. easy to store

long storage time;

[3]

- (g) Embryonic stem cells have been known to express a specific set of genes that enable them to behave like stem cells, but these genes are turned off in normal somatic cells. Researchers have discovered that they can convert human fibroblast cells into embryonic stem cells by switching on the specific set of genes. These cells have been cultured in labs all over the world, and they are termed induced pluripotent stem cells (iPSCs).

- (i) Describe the characteristics of an induced pluripotent stem cell (iPSC).

1. An iPSC cell is an undifferentiated / unspecialised cell

capable of undergoing proliferation / mitotic division / cell division / self-renewal;

OR

An iPSC is an undifferentiated / unspecialised cell lacking tissue-specific structures;

2. Can differentiate into all cell types in an organism except extraembryonic tissue

(hence cannot form an entire organism / foetus on its own);

3. An iPSC can differentiate to produce specialised cells

as a result of differential switching on of genes / differential gene expression upon receiving molecular signals;

4. An iPSC has long term self-renewal capacity due to presence of telomerase,

ensures maintenance of telomere length / replacement of telomeres, enabling cells to overcome the Hayflick limit (and continue cell division / mitotic division / proliferation);

[3]

Max 3 marking points

- (ii) Suggest an ethical concern regarding the use of embryonic stem cells which is no longer relevant with the development of iPSCs.

1. No destruction of embryos with use of iPSC

hence not regarded as killing a life due to some views that embryos are living organisms;

2. Some object to extracting stem cells from an embryo to make replacement body cells

is treating embryo as just a source of spare parts;

[1]

Any 1 marking point above

The histone tails of different regions of chromatin are covalently modified during the formation of the human fibroblast cell. These modifications are added and removed by enzymes. The main effect of modifying histone tails is to attract specific proteins to a stretch of chromatin.

Fig. 1.5 shows the action of one of these enzymes, histone methyl transferase.

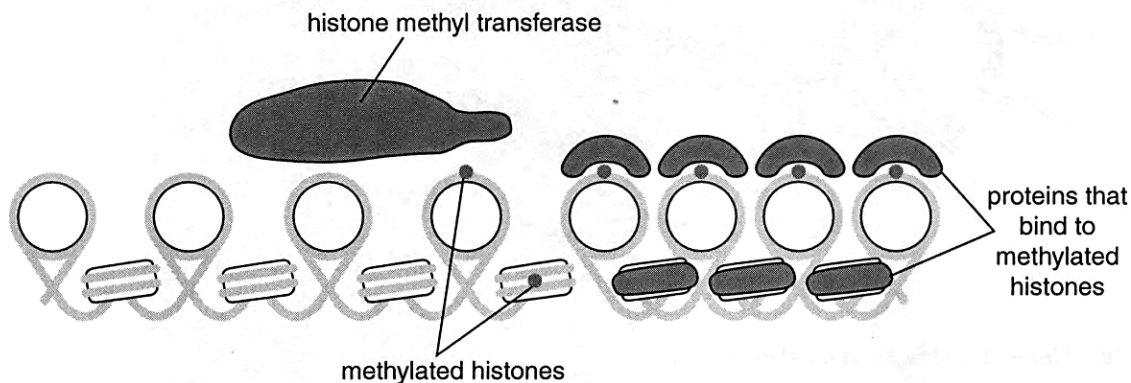


Fig. 1.5

(iii) Suggest why histone methylation occurs over large areas of chromatin during the formation of the human fibroblast cell.

1. **only specific group of genes are transcribed during formation of human fibroblast cell to produce specific proteins / enzymes to perform specific function;**

2. **of the specific group, only a few genes are expressed each time to respond to changes in external environment / needs of cell;**

3. **regions with genes that are not expressed will be tightly coiled through histone methylation,**

many regions on chromatin are non-coding sequence thus are not transcribed & will be tightly coiled;

[3]

[Total: 30]

- 2 Today, viruses are considered an exception to the cell theory which states that the simplest units of life are cells. Viruses have also been referred to as “organisms at the edge of life”.

(a) State **one** characteristic of viruses that may classify them as being

(i) living,

1. able to reproduce / replicate with the aid of a host cell

2. possess nucleic acids containing genetic information for reproduction

3. able to obtain and use energy

4. able to respond to environment e.g. by switching replication cycles

5. able to evolve and mutate

[1]

Any 1 marking point

(ii) non-living.

1. are acellular

2. lack cytoplasm and metabolic machinery / organelles

e.g. nucleus, mitochondria, ribosomes

3. cannot undergo metabolism

e.g. cellular respiration / synthesis of organic compounds

4. cannot replicate independently

are obligate parasites / dependent on host cell for metabolic machinery to replicate and synthesise new viral components

5. unable to generate or store energy

derive ATP from host cell

6. contains only 1 type of nucleic acid

either DNA or RNA but not both

[1]

Any 1 marking point

Coronavirus disease 2019 (COVID-19) is an infectious disease caused by a novel coronavirus, severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

Fig. 2.1 shows the mechanism used by SARS-CoV-2 to infect the cells in the airways of human hosts.

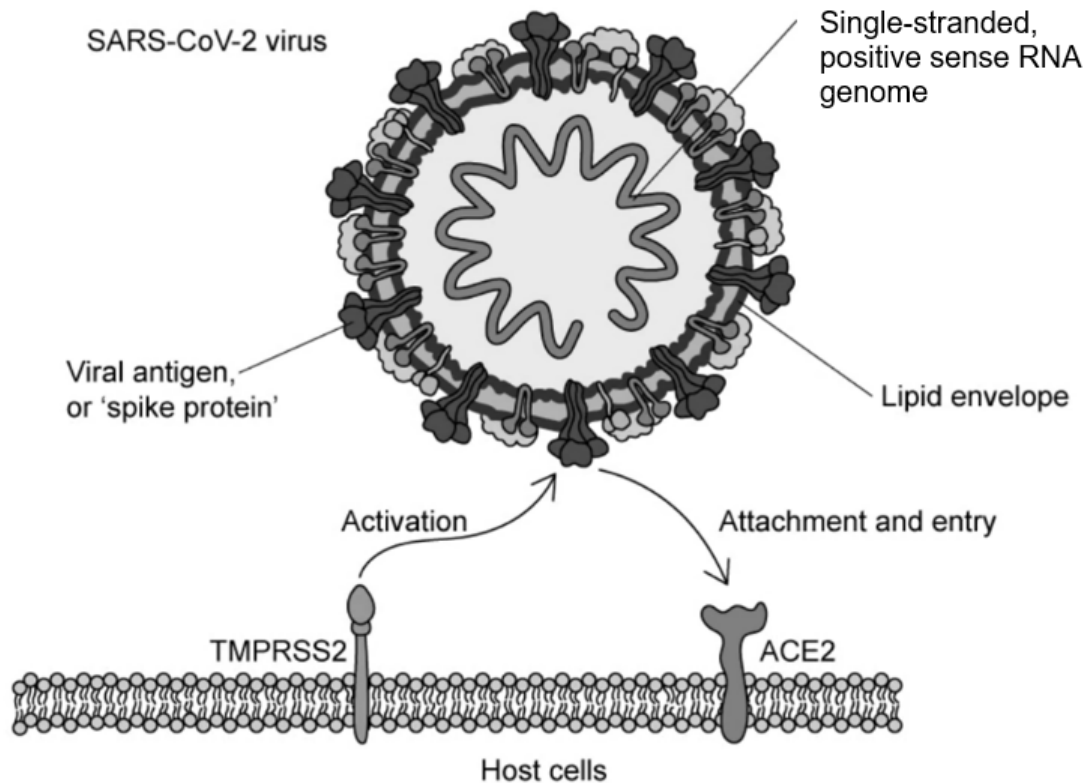


Fig. 2.1

(b) With reference to Fig. 2.1, suggests how SARS-CoV-2 infects human cells.

1. **viral antigen / spike protein binds TMPRSS2 (on host cell membrane)**
via complementary fit;
2. **resulting in a conformational change to viral antigen**
activating it;
3. **activated viral antigen is now complementary to ACE2;**
4. **allowing binding of viral antigen to result in virus entering by receptor mediated endocytosis;**

[3]

A drug currently being trialled for the treatment of COVID-19 is Nafamostat, a drug that is currently licensed for the treatment of other conditions.

During one trial, scientists investigated the effect of Nafamostat on the activity of TMPRSS2, and the results are shown in Fig. 2.2 below.

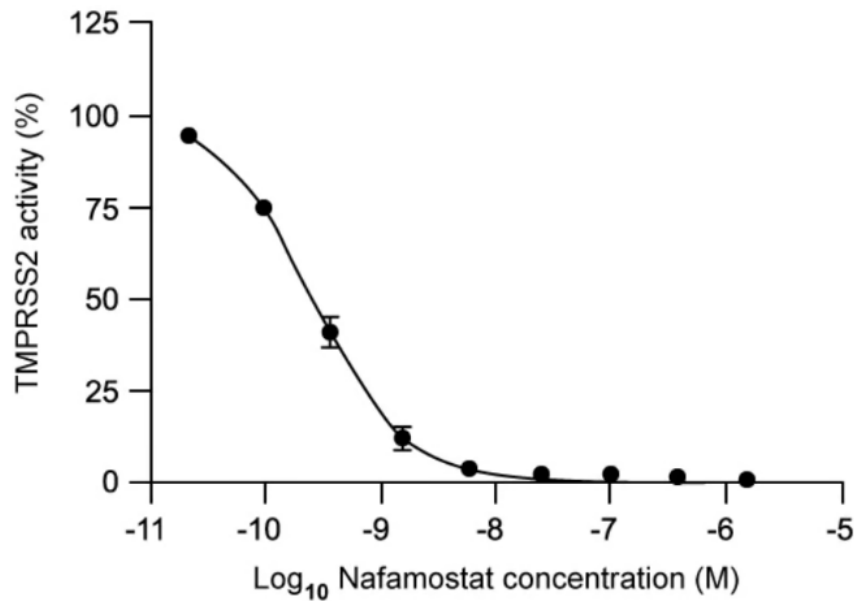


Fig. 2.2

- (c) With reference to Fig. 2.1 and and Fig. 2.2, suggest how Nafamostat might function as a treatment for COVID-19.

1. as Nafamostat concentration increases from $10^{-10.5}$ to 10^{-7} M, @ 10^{-11}

.....
 TMPRSS2 activity decreases (steeply) from 95% to 0%; @ 100%

2. Nafamostat is (structurally similar to viral antigen / complementary in shape to the TMPRSS2)

.....
 able to bind and block / inhibit TMPRSS2 from binding to viral antigen;

3. virus antigen is not activated

.....
 hence cannot bind ACE2 to enter host cell;

[3]

Fig. 2.3 shows the reproductive cycle of the coronavirus SARS-CoV-2.

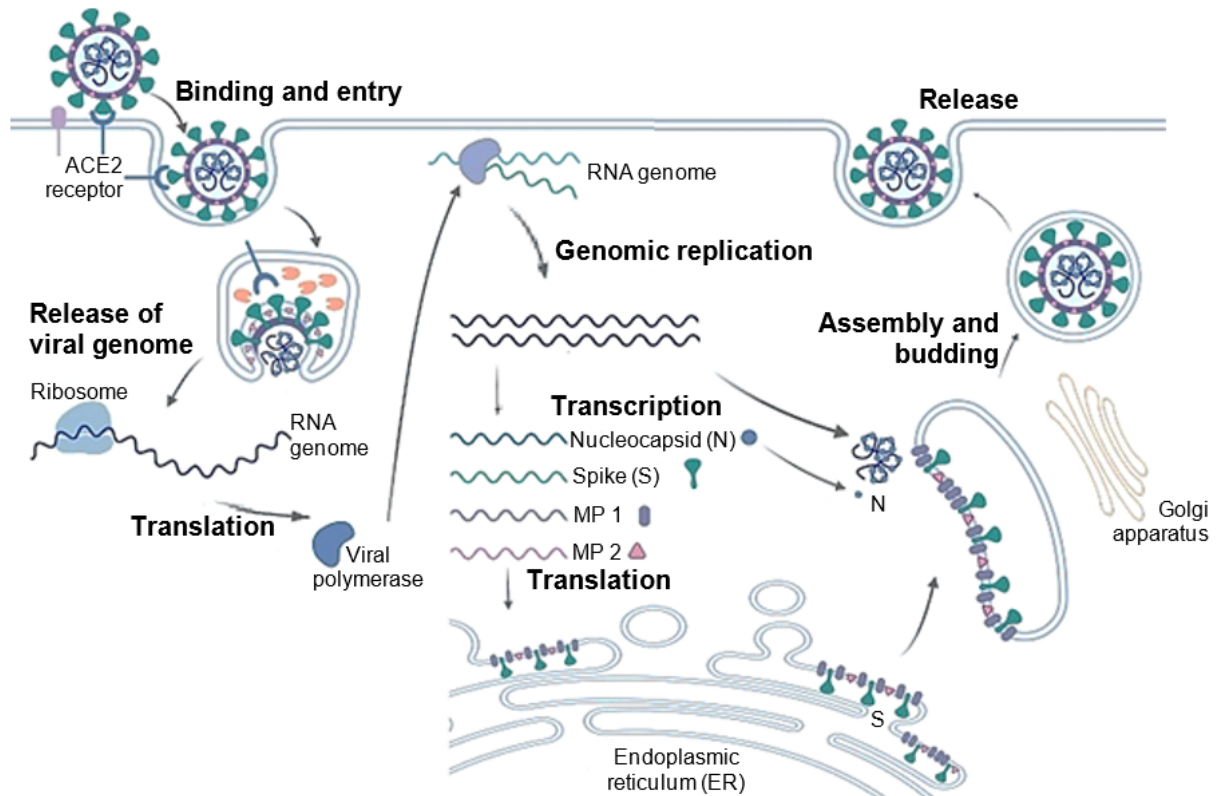


Fig. 2.3

(d) Compare between the reproductive cycles of SARS-CoV-2 and the influenza virus.

1. SARS-CoV-2: spike protein binds to ACE2 receptor

Influenza virus: haemagglutinin binds to sialic acid containing receptor;

2. SARS-CoV-2: exit by exocytosis

Influenza virus: exit by budding;

3. SARS-CoV-2: positive sense RNA genome

Influenza virus: negative sense RNA genome;

4. both require viral polymerase / RNA dependent RNA polymerase for genome replication

5. both enter host cell by receptor mediated endocytosis;

[2]

max. 1 similarity
max. 1 difference

® SARS-CoV-2 assemble viral proteins in rER while influenza assemble at cell surface membrane, as HA and NA are inserted into rER membrane before being transported to cell surface membrane.

[Total: 10]

- 3 Table 3.1 shows dengue incidences and precipitation pattern in a tropical country from 2013 to 2017.

Table 3.1

Year	Average annual rainfall/mm	Number of cases of Dengue Fever	Average Maximum Temperature/°C	Average Minimum Temperature /°C
2013	665	20270	32.3	23.2
2014	585	19902	33.9	23.2
2015	578	18910	33.8	24.2
2016	695	4260	34.8	24.5
2017	564	2145	36.2	25.0

- (a) With reference to Table. 3.1, comment on the relationship between dengue fever prevalence and climate factors.

1. QV Number of cases from highest to lowest in relation to decreased rainfall

e.g. 20270 in 2013 to 2145 in 2017, 665mm to 564mm;

2. QV Number of cases from highest to lowest in relation to increased maximum and minimum temperature

e.g. 20270 in 2013 to 2145 in 2017, 32.3 °C to 36.2 °C with 23.2 °C to 25.0 °C;

3. High annual rainfall increases number of breeding sites / stagnant water

for A. egypti to lay eggs;

4. High maximum and minimum temperature exceeded the temperature range for A. egypti

Decreased development and reproduction rate;

[4]

To reduce the transmission of dengue fever through decreasing vector numbers, scientists used high doses of radiation on male *A. aegypti* before releasing large numbers into the wild.

- (b) Suggest how releasing the treated male mosquitoes may reduce transmission of dengue fever.

1. Radiation destroyed reproductive cells of male mosquitoes

leading to sterility;

2. Mating do not result in fertilization

no production of offspring;

[2]

Recently a new method was developed to control *A. aegypti*. Scientists produced transgenic males carrying a 'lethal gene' which kills their offspring before they can reproduce.

The scientists released transgenic males every week in one area of a city in Brazil. At regular intervals they determined the number of *A. aegypti* per km² in the area where transgenic males were released and in a control area where no transgenic males were released.

Fig. 3.1 shows their results.

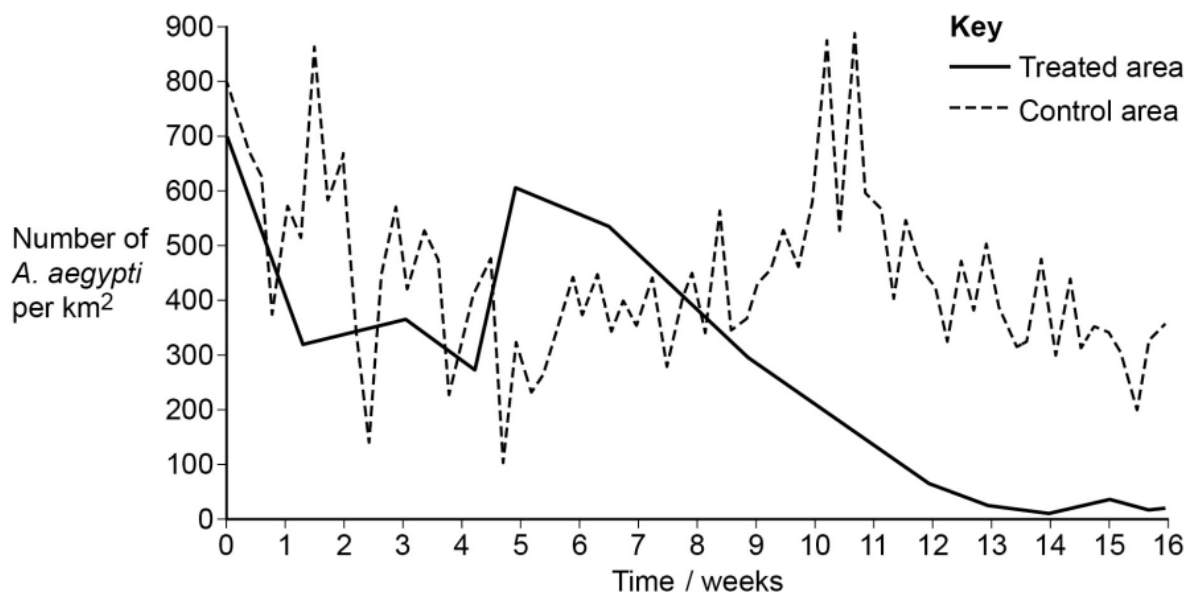


Fig. 3.1

- (c) (i) With reference to Fig. 3.1, conclude whether the release of transgenic males successfully reduced the vector population.

1. successful from Week 0 to 4.5

Numbers of mosquitos decrease from 700 and lowest at 280 per km²;

2. not successful from Week 4.5 to Week 8.5

Number of mosquitoes increased to 600 per km², higher than control area;

3. Successful from Week 9 onwards

Number of mosquitoes in treated area consistently lower at 280 compared to 380 per km² in control area;

OR

4. Successful, decreased from Week 0 to 16

Number of mosquitoes from 700 per km² to less than 50 per km²;

[3]

Marking points 1 +2 AND 3 / 4 for full credit

- (ii) Suggest why the scientists released more transgenic males every week.

Short life span / ensure sufficient transgenic male mosquitoes to ensure reproductive competitiveness;

AVP

[1]

[Total: 10]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

You answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

- 4 (a) DNA can be replicated *in vivo* (in cells) or *in vitro* (outside cells) by processes like the polymerase chain reaction (PCR).

Compare the process of DNA replication in cells with PCR.

[10]

Feature	DNA replication in cells	PCR
1. end replication problem	occurs in cells resulting in 3' overhang in template strand / shortening of daughter strand	does not occur, 3' end of template strand is completely replicated
2. nature of primers	RNA primers synthesised by primase	DNA primers artificially synthesised and added to reaction mixture
3. number of types of primers	many types, each leading and lagging strand with own primer complementary to different sections of the template strand	2 types, forward and reverse, complementary to sequences flanking the target sequence
4. fate of primers	removed by DNA polymerase and replaced with DNA nucleotides	not removed, forms part of the replicated DNA
5. DNA polymerases	more than 1 type, 1 to elongate daughter strand, 1 to remove primer and fill up the gap	1 type, Taq polymerase
6. template	entire parental strand is replicated	only target sequence (segment of template strand) is replicated
7. starting point of replication	multiple origins of replication	flanking sequence that primers are complementary to
8. separation of template strand	hydrogen bonds between complementary bases broken by helicase	hydrogen bonds between complementary bases broken by increase in kinetic energy due to increase in temperature

Feature	DNA replication in cells	PCR
9. product	shorter than parent strand at 5' end (by 1 primer length)	mostly length of target sequence with those daughter strands using original DNA as template being longer
10. product	leading strand synthesised continuously, lagging strand synthesised discontinuously in Okazaki fragments	all daughter strands synthesised continuously
11. product	Double stranded DNA molecule	
12. mode of replication	Semi-conservative from 5' to 3'	
13. Nature of DNA polymerase	Require primer to provide a free 3'OH end for elongation to synthesize the new DNA strand	
14. Original Template	Requires DNA as the template	
15. Enzymes involved in formation of phosphodiester bonds	Requires DNA polymerase	

16. AVP – Differences

17. AVP – Similarities

QWC: include at least 1 similarity and differences

- (b) Both DNA replication in cells and PCR may harbour mutations resulting in changes to DNA base sequences.

Discuss the impact of mutations to organisms.

[15]

1. results in creation of new alleles
2. increase genetic diversity in gene pool for natural selection
3. if new allele code for favourable phenotype, individual is selected for
4. e.g. antibiotic resistance in bacteria / melanism in moth during industrialisation
5. if new allele code for detrimental / unfavourable phenotype, individual is selected against
6. e.g. CFTR / gain of function mutation in protooncogene / loss in function mutation in tumour suppressor gene resulting in cancer
7. imprecise joining of V(D)J segments (in hypervariable region) during somatic recombination in B (and T) cells maturation
8. results in wide repertoire of B (and T) cell receptors that can bind to wide variety of antigens

9. somatic hypermutation during clonal expansion / differentiation in plasma cells
10. results in variety of immunoglobulin with different affinity to specific antigen
11. allowing for affinity maturation which selects plasma cells that can produce antibodies that binds well to specific antigen
12. increasing efficiency of immune response in secondary responses
13. may result in B cells that produce antibodies with self-reactivity resulting in auto-immunity
14. antigenic drift in virus results in changes to antigenic epitope allowing virus to evade detection by memory B cells
15. antigenic shift in virus results in recombination of genetic material from different strains giving rise to ability to infect new host type (cross species barrier)
16. Chromosomal mutations leading to changes in structure or numbers of chromosomes to result in speciation or disease
17. E.g. of polyploidy in maize leading to sympatric speciation
18. Gene mutations leading to changes in protein structure resulting in disease
19. E.g. sickle cell anaemia due to change of amino acid from glutamate to valine to result in hydrophobic protrusion which fits into hydrophobic pocket during low partial pressure of oxygen resulting in sickle shaped red blood cells
20. E.g. Down syndrome, aneuploidy of the 21st chromosome where there are 3 chromosome 21 resulting in a trisomy 21 leading to intellectual/ physical disabilities
21. Isolated populations accumulate mutations independently to result in reproductive barriers, resulting in genetic divergence to result in speciation
22. AVP

QWC: at least 2 mutations linked to relevant impact in 2 different areas

Examiner's Report:

Many candidates incorrectly went on to describe in a detailed manner the different types of mutations without relating to the impact of the organism. Answers which were awarded credit were frequently in the area of mutations to proto-oncogene, tumour suppressor genes and β -globin gene. Candidates who mentioned natural selection failed to identify correctly the impact of the selection pressure on the organisms and vaguely mentioned that favourable alleles are inherited by the offspring without relating to changes in gene pool.

[Total: 25]

- 5 (a) Both eukaryotic cells and bacterial cells are able to divide to give daughter cells. Eukaryotic cells can divide by mitosis to produce cells for growth and repair or meiosis to produce gametes. Bacterial cells, however, can only divide by binary fission to result in clones.

Compare the process of mitosis in eukaryotic cells and binary fission in bacterial cells.

[10]

Feature	Eukaryotic cell	Bacterial cell
1. Location of chromosome	each chromosome is free floating in cytosol	each DNA strand of bacterial chromosome attaches to cell surface membrane
2. occurrence of chromosome after replication	replicated DNA are joined at centromere forming 2 sister chromatids after replication	chromosomes exist as 2 separate chromosomes after replication
3. arrangement of chromosomes	chromosomes are lined up in single row at metaphase plate during metaphase	chromosomes are attached to cell surface membrane adjacent to each other
4. separation of chromosomes	sister chromatids are pulled apart to opposite poles of the cell by kinetochore spindle fibres during anaphase	daughter chromosomes are separated as cell wall material is deposited between the points of attachment
5. location of chromosomes at the end of division	nuclear envelope forms around daughter chromosomes at each pole forming new nuclei	daughter chromosomes are in nucleoid region in cytoplasm
6. division of cell	cleavage furrow (for animal cells) or cell plate (for plant cells) forms at equator to separate the parent cell occurs concurrently with telophase (in mitosis)	formation of septum between 2 daughter chromosome separates the parent cell into 2 daughter cells
7. Duration of cellular division	Longer due to larger genome and presence of organelles	Shorter with a smaller genome and no organelles
8. Ploidy of parent cell	Diploid with 2 sets of homologous chromosomes	Haploid with 1 circular chromosome
9. DNA replication	Both require chromosomes to be replicated <u>before</u> separation of DNA into new daughter cells.	
10. Daughter cells	Have the same number of chromosomes and genetic material as the parent cell / genetically identical to each other (and to parent cell)	

Max: 9 marking points for comparative statements
QWC: awarded for at least 1 similarity AND 1 difference

- (b) Discuss how genetically varied cells can arise despite binary fission being the only mode of cell division for bacterial cells. [15]
1. mutation resulting in change in nucleotide sequence in DNA
 2. can be spontaneous during DNA replication
 3. or induced due to exposure to mutagens / ionising radiation
 4. resulting in changes in base structure or pairing properties of the bases
 5. error in DNA replication where wrong bases are paired
 6. due to error in proof reading by DNA polymerase
 7. transformation – competent bacteria cells take up naked DNA from the environment
 8. and undergo successful homologous recombination
 9. conjugation between F⁺ and F⁻ cells
 10. where F plasmid from F⁺ cell is transferred to F⁻ cell via conjugation bridge
 11. transduction where bacteriophages transfers genetic material from 1 host to the next
 12. generalised transduction involving lytic phages which package and transfer random fragments of lysed bacterial host chromosome to next host
 13. specialised transduction involving temperate phages which package and transfer specific fragments of lysed bacterial host chromosome adjacent to prophage to next host
 14. successful gene transfer between bacteria result in acquisition of new alleles
 15. State e.g. of genes acquired from gene transfer processes e.g. genes coding for antibiotic resistance / enzymes to use new metabolites / etc.

QWC: at least 2 different modes

[Total: 25]